Problem Set for 12-May and 15-May

The ambient field is $\mathbb{R}$.

Exercise 1 Determine whether the following are inner product spaces (简要回答即可).

1. V is the vector space continuous functions over $\mathbb{R}$, and the pairing is

$$(f,g) \mapsto \int_0^1 f(x)g(x)\,\mathrm{d}\,x.$$

> *不是内积空间. 存在 $f \neq 0$ 使得 $(f,f)=0$.*

2. $V = \mathbb{R}[x]$, and the pairing is

$$(f,g) \mapsto \int_{-\infty}^{\infty} f(x)g(x)e^{-x^2}\,\mathrm{d}\,x.$$

> *是内积空间. 显然 (f,g) 是对称双线性型. 对任意 f 总有 $(f,f)<\infty$ (积分收敛). 从而 $(f,g)=\frac{(f+g,f+g)-(f,f)-(g,g)}{2}$ 良定义.*

3. $V = \mathbb{R}^{n\times n}$, and the pairing is

$$(A,B) \mapsto \det(A^T \cdot B).$$

> *不是内积空间 (除去 $n=1$ 的平凡情况). 存在 $A \neq O$ 使得 $\det(A^T A)=0$.*

4. (**optional**) By axiom of choice, any $\mathbb{R}$-vector space can be endowed with an inner product structure.

> *取定义线性空间 V 的一组基 $\mathscr{B}$, 则任意 $x \in V$ 能唯一地写作**有限和** $\sum \lambda_i x_i$. 定义正定二次型 $(\sum \lambda_i x_i, \sum \lambda_i x_i) = \sum \lambda_i^2$, 并由此定义内积即可.*

Exercise 2 Show that every real matrix is similar to a block diagonal matrix, wherein the candidates of the blocks are

1. the Jordan form of a real eigenvalue;
2. the block of the form

$$\begin{bmatrix} H_{r,\theta} & Q & & & \\ & H_{r,\theta} & Q & & \\ & & \ddots & \ddots & \\ & & & H_{r,\theta} & Q \\ & & & & H_{r,\theta} \end{bmatrix},$$

where

1. $H_{r,\theta} = \begin{bmatrix} r\cos\theta & -r\sin\theta \\ r\sin\theta & r\cos\theta \end{bmatrix}$, and
2. $Q = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$.

> *对实矩阵在 $\mathbb{C}$ 中的 Jordan 型, 非实数 Jordan 块成对出现. 下直接验证*
>
> $$\begin{bmatrix} H_{r,\theta} & Q & & & \\ & H_{r,\theta} & Q & & \\ & & \ddots & \ddots & \\ & & & H_{r,\theta} & Q \\ & & & & H_{r,\theta} \end{bmatrix} \sim \begin{bmatrix} J_n(re^{i\theta}) & O \\ O & J_n(re^{-i\theta}) \end{bmatrix}$$
>
> *记 L 为左侧矩阵, R 为右侧矩阵. 依照分块对角矩阵的行列式,*
>
> $$\det(\lambda I - L) = (\det(\lambda I - H_{r,\theta}))^n = (x^2 - 2xr\cos\theta + r^2)^n = \det(\lambda I - R).$$
>
> *任取 $\mu \in \{re^{\pm i\theta}\}$, 使用待定系数计算得 $\dim\ker(\mu I - L) = 1$. 例如取 $(\mu I - L) \cdot v = 0$,*
>
> - *若 v 前两项全为 0, 则 v 的第三与第四项为 0, 归纳得 $v = 0$;*
> - *若前两项非零, 则比值确定. 取定 v 的前两项, 则 v 的第三与第四项唯一确定.*
>
> *以上说明 L 在 $\mathbb{C}$ 上的标准型只有两个 Jordan 块, 因此只能是 R 的形式.*

Exercise 2.5 Show that every real matrix is a product of two real symmetric matrices.

> *记 T 是 $\{0,1\}$-矩阵, 其中 $T_{i,j} = 1$ 当且仅当 $i + j = n + 1$.*
>
> *将任意 X 写作上题中的标准型 $P^{-1}LP$, 则 $L = LT \cdot T$ 是对称矩阵的乘积. 因此*
>
> $$X = P^{-1}LTP^{-1,T} \cdot P^{T}LP$$

> 是两个对称矩阵的乘积.

Exercise 3 Let $(V,(-,-))$ be a finite dimensional real inner product space. For unit vector v_0 (i.e., $(v_0,v_0)=1$), we define the reflection

$$\varphi : V \to V, \quad x \mapsto x - 2(x,v_0)\cdot v_0.$$

Show that for isometric transform ψ (i.e., $(\psi(u),\psi(v))=(u,v)$), 1 is an eigenvalue for ψ or $\psi\circ\varphi$. Overall, $1\in\sigma(\psi)\cup\sigma(\psi\circ\varphi)$.

> *本题先猜后证. 关于 x 的构造, 可顺着目标式 $\varphi(x)=\psi(x)$ 画图猜测.*
>
> *若 1 不是 ψ 的特征值, 则 $(\mathrm{id}-\psi)$ 是同构. 此时, 存在 x 使得 $(\mathrm{id}-\psi)(x)=v_0$. 从几何角度看, $x-\psi(x)$ 与 v_0 同向, 且 ψ 等距, 则应有 $(x+\psi(x),v_0)=0$. 计算得*
>
> $$(x+\psi(x),v_0)=(x+\psi(x),x-\psi(x))=(x,x)-(\psi(x),\psi(x))=0.$$
>
> *此时*
>
> $$\begin{aligned}\varphi(x)&=x-(2x,v_0)v_0\\&=x-(x-\psi(x),v_0)v_0\\&=x-(v_0,v_0)v_0\\&=x-v_0\\&=x-(x-\psi(x))\\&=\psi(x).\end{aligned}$$
>
> *由 $\varphi^2=\mathrm{id}$, 得 $1\in\sigma(\psi\circ\varphi)$.*

Exercise 4 Set $V:=\mathbb{R}[x]$ and $V_0:=\{f\in\mathbb{R}[x]\mid f(0)=f(1)\}$.

1. Prove that $V\times V\to\mathbb{R},\quad (f,g)\mapsto\int_0^1 f(x)g(x)\,\mathrm{d}\,x$ is an inner product.
2. Set $\mathscr{D}:V_0\to V,\quad f(x)\mapsto f'(x)$. Find $\dim\ker(\mathscr{D})$ and $\dim\operatorname{coker}(\mathscr{D})$.
3. Define the inner product restricted on the subspace

$$(\cdot,\cdot)_0 : V_0\times V_0\to\mathbb{R},\quad (f,g)\mapsto\int_0^1 f(x)g(x)\,\mathrm{d}\,x.$$

   Is there any linear map $\mathscr{D}^*: V\to V_0$ such that for any $h\in V_0$ and $g\in V$,

$$(\mathscr{D}^*g,h)_0=(g,\mathscr{D}h)?$$

内积的对称双线性性是显然的, 关键是说明 $(f,f)=0 \implies (f=0)$. *若多项式在* $(0,1)$ *上恒零, 则自然是零多项式.*

记 $\mathscr{D}=D\circ i$, *其中* $i:V_0\to V$ *是包含映射,* D *是* V *上的导数.*

1. *由* i *是单射, 故* $\dim\ker Di\le\dim\ker D=1$, *得* $\dim\ker Di\in\{0,1\}$. *由常多项式属于* V_0, *取* 1.
2. *由* D *是满射, 故* $\dim\operatorname{coker} Di\le\dim\operatorname{coker} i=1$, *得* $\dim\operatorname{coker} Di\in\{0,1\}$. *由常多项式不属于* $\operatorname{im}(Di)$, *取* 1.

不存在. 下使用反证法. 若 $\mathscr{D}^*$ *存在, 对任意* h $(h(0)=h(1)=0)$ *与* $g\in V$, *总有*

$$(\mathscr{D}^*g,h)_0=(g,\mathscr{D}h)=(-g',h).$$

取 $V_1=\{x(x-1)f\mid f\in V\}$ *为以上* h *所在的线性空间, 因此* $(\mathscr{D}^*g+g')\in V_1^{\perp}$. *对任意* $\varphi\in V_1^{\perp}$, *有*

$$0=(\varphi,x(x-1)\varphi)=\int_0^1 x(x-1)\varphi^2\,\mathrm{d}\,x.$$

因此 $\varphi=0$. *这说明* $V_1^{\perp}=0$, *从而* $(\mathscr{D}^*g+g')=0$. *这与* $\operatorname{im}(\mathscr{D}^*)\subseteq V_0$ *矛盾.*

Exercise 5 Let $(V,(,))$ be an inner product space, where V is not necessary finite dimensional.

- Say $\varphi^\star$ is the adjoint of $\varphi\in\operatorname{Hom}_{\mathbb{R}}(V,V)$, provided $(\varphi^\star(x),y)=(x,\varphi(y))$ for arbitrary $x,y\in V$.
- Let U be a subspace of V. Set

$$U^{\perp}:=\{v\in V\mid (u,v)=0,\forall u\in U\}=\bigcap_{u\in U}\ker((u,-))$$

Now we assume that φ is invertible, and $\varphi^\star$ exists.

1. Show that $\varphi^\star$ is injective, and $(\operatorname{im}(\varphi^\star))^{\perp}=0$.

即证 $[\varphi^\star(y)=0] \implies [y=0]$. *由* φ *可逆, 取原像* $x=\varphi^{-1}(y)$, *得*

$$0=(\varphi^\star(y),x)=(y,\varphi(x))=(y,y).$$

由内积空间定义, $y=0$.

再证 $z\in\operatorname{im}(\varphi^\star)^{\perp}$ *必为* 0. *此时* $(\varphi^\star(-),z)=(-,\varphi(z))$ *是零映射, 因此* $\varphi(z)=0$, *故* $z=0$.

2. Show that $\varphi^\star$ is surjective $\implies (\varphi^{-1})^\star$ exists and $(\varphi^{-1})^\star = (\varphi^\star)^{-1}$.

> *结合上一问, 题设是 $\varphi^\star$ 为同构. 此时*
>
> $$(x, \varphi^{-1}(y)) = ((\varphi^\star)(\varphi^\star)^{-1}(x), \varphi^{-1}(y)) = ((\varphi^\star)^{-1}(x), \varphi(\varphi^{-1}(y))) = ((\varphi^\star)^{-1}(x), y).$$
>
> *这一构造表明 $(\varphi^\star)^{-1} = (\varphi^{-1})^\star$.*

3. Show that $(\varphi^{-1})^\star$ exists $\implies \varphi^\star$ is invertible and $(\varphi^{-1})^\star = (\varphi^\star)^{-1}$.

> *若 $(\varphi^{-1})^\star$ 存在, 则*
>
> $$((\varphi^{-1})^\star(\varphi^\star)(x), y) = (\varphi^\star(x), \varphi^{-1}(y)) = (x, \varphi(\varphi^{-1}(y))) = (x, y).$$
>
> *这说明 $(\varphi^{-1})^\star\varphi^\star$ 是恒等映射. 类似的计算 (调换位置) 表明 $\varphi^\star(\varphi^{-1})^\star$ 是恒等映射, 从而 $\varphi^\star$ 与 $(\varphi^{-1})^\star$ 是互逆的.*

4. Let $V = \mathbb{R}[x]$ and $(f, g) := \sum_{n\geq 0} f_i g_i$, where $h = \sum h_i \cdot x^i$. Show that
   1. $(V, (,))$ is an inner product space;
   2. $L : f \mapsto \frac{f - f(0)}{x}$ is the linear map of moving left. Show that $(\mathrm{id} + L)$ is invertible;
   3. Show that $\varphi := (\mathrm{id} + L)^{-1}$ has no adjoint $\varphi^\star$.

> *这都是直接的验证. 对于最后一点, 若 $((\mathrm{id} + L)^{-1})^\star$ 存在, 则 $(\mathrm{id} + L)^\star$ 是可逆的映射. 直接的计算表明 $(\mathrm{id} + L)^\star = (\mathrm{id} + R)$ 非满, 因为 $x \in \mathrm{im}(\mathrm{id} + R)$ 必然满足 $\sum x_i = 0$.*

Exercise 6 (**Optional, very interesting**). Find the dimension and inertia of the following symmetric bilinear form, and verify the invariant group action.

1. $V = \{X \in \mathbb{R}^{2\times 2} \mid \mathrm{tr}(X) = 0\}$, and the pairing is $(A, B) \mapsto \mathrm{tr}(A \cdot B)$. Moreover, for $g \in \mathrm{GL}_2(\mathbb{R})$,the pairing is invariant under $X \mapsto g \cdot X \cdot g^{-1}$.

> ***Step 0*** *$\dim V = 3$.*
>
> ***Step 1*** *The pairing is linear (trivial) and symmetric, since $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.*
>
> ***Step 2*** *Let $(A_1, A_2, A_3) := (E_{1,2}, E_{2,1}, E_{1,1} - E_{2,2})$ be a basis of V, the associated Gram matrix is the 3×3 symmetric matrix $(\mathrm{tr}(A_i A_j))_{3\times 3}$ is $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. The inertia is $(+ + -)$.*

> ***Step 3*** *Since* $\mathrm{tr}((gXg^{-1})\cdot(gYg^{-1}))=\mathrm{tr}(XY)$, *pairing is invariant under such group action.*

2. $V=\mathbb{R}^{2\times 2}$, and the pairing is $(A,B)\mapsto \mathbf{tr}(\mathbf{adj}(A)\cdot B)$. Moreover, for $(g,h)\in \mathrm{SL}_2(\mathbb{R})\times\mathrm{SL}_2(\mathbb{R})$, the pairing is invariant under $X\mapsto g\cdot X\cdot h^{-1}$.

> $\dim V=4$. 基 $(E_{1,1}\pm E_{2,2},E_{1,2}\pm E_{2,1})$ 对应惯性指数 $(\pm,\pm)$. 对 $g\in\mathrm{SL}_2$, $\mathrm{adj}(g)=g^{-1}$.
>
> $$\mathrm{tr}(\mathrm{adj}(gXh^{-1})gYh^{-1})=\mathrm{tr}(h\mathrm{adj}(X)g^{-1}gYh^{-1})=\mathrm{tr}(\mathrm{adj}(X)Y).$$

3. $V=\left\{\begin{pmatrix} a & ib\\ ic & \overline{a}\end{pmatrix}\mid a\in\mathbb{C};b,c\in\mathbb{R}\right\}$, the pairing is $(X,Y)\mapsto \mathbf{Re}(\mathbf{tr}(X\cdot\overline{Y}))$. Moreover, for $g\in\mathrm{GL}_2(\mathbb{C})$, the pairing is invariant under $X\mapsto g\cdot X\cdot(\overline{g})^{-1}$.

> $\dim V=4$. 基 $(I,i(E_{1,1}-E_{2,2}),i(E_{1,2}\pm E_{2,1}))$ 对应惯性指数 $(+,+,\pm)$. 对 $g\in\mathrm{GL}_2(\mathbb{C})$,
>
> $$\mathrm{Re}(\mathrm{tr}(gX\overline{g}^{-1}\cdot\overline{gY\overline{g}^{-1}}))=\mathrm{Re}(\mathrm{tr}(X\cdot\overline{Y})).$$

4. (**Optional**, if you know tensor product). Let $U\wedge U$ be the quotient space of $U\otimes U$ by the relation $(a\otimes b+b\otimes a=0)$. In particular, $\dim(U\wedge U)=\binom{\dim U}{2}$. Let $V=\mathbb{R}^4\wedge\mathbb{R}^4$ be the 6-dimensional vector space with basis $\{e_i\wedge e_j\}_{1\le i<j\le 4}$, and the pairing is $(\lambda,\mu)\mapsto\lambda\wedge\mu$. Here $\bigwedge_{i=1}^4\mathbb{R}^4$ is one-dimensional with basis $\{e_1\wedge e_2\wedge e_3\wedge e_4\}$. Moreover, for $g\in\mathrm{SL}_4(\mathbb{R})$, the pairing is invariant under $\mu\mapsto g\cdot\mu$. Here $g\cdot(\sum c_{i,j}e_i\wedge e_j):=\sum c_{i,j}(g\cdot e_i)\wedge(g\cdot e_j)$.

> $\dim V=6$. 基
>
> $$(e_1\wedge e_2\pm e_3\wedge e_4,e_1\wedge e_3\pm e_2\wedge e_4,e_1\wedge e_4\pm e_2\wedge e_3)$$
>
> 对应惯性指数 $(\pm,\mp,\pm)$. 对 $g\in\mathrm{SL}_4(\mathbb{R})$,
>
> $$\det(g\cdot\sum u_i\wedge v_i,g\cdot\sum x_j\wedge y_j)=\sum_{i,j}\det(gu_i\wedge gv_i\wedge gx_j\wedge gy_j).$$
>
> 考虑 $\det ge_1\wedge ge_2\wedge ge_3\wedge ge_4=(\det g)\cdot(\det e_1\wedge e_2\wedge e_3\wedge e_4)$, 上式 g 可去.

The above calculation provides the elementary construction of low dimensional spin groups, that is, the double covering of special orthogonal groups. One learns from these construction that

1. $\mathrm{SL}_2(\mathbb{R})/\{\pm I\} \simeq \mathrm{SO}(2,1)$, and $\mathrm{SL}_2(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}_3(\mathbb{C})$;
2. $(\mathrm{SL}_2(\mathbb{R}))^2/\{\pm(I,I)\} \simeq \mathrm{SO}(2,2)$, and $(\mathrm{SL}_2(\mathbb{C}))^2/\{\pm(I,I)\} \simeq \mathrm{SO}_4(\mathbb{C})$;
3. $\mathrm{SL}_2(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}(3,1)$;
4. $\mathrm{SL}_4(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}_6(\mathbb{C})$, and $\mathrm{SL}_4(\mathbb{R}) \simeq \mathrm{SO}(3,3)$.

As corollaries, $\mathrm{SO}_4(\mathbb{C})$ is a double cover of $\mathrm{SO}_3(\mathbb{C})$, and the group isomorphism $\mathrm{SO}(3,1) \simeq \mathrm{SO}_3(\mathbb{C})$ is a famous coincidence in quantum physics.